MechAgent engineering report

Status: VERIFIED: a design passed every check, the convergence study and validation

Units: mm, N, MPa, kg. Every number comes from deterministic tools, the LLM only read the request and chose between allowed actions (marked [llm] in the trace).

1. Request

Original text:

> Shelf bracket on a vertical wall. Global frame: z points up, the wall is the plane x = 0 and the bracket is on the x > 0 side.
>
> Mounting: two bolt holes of 16 mm (M15 bolts) at (0, -40, 200) mm and (0, 40, 200) mm, bolt axis (1, 0, 0).
>
> Load: a pin goes through a 20 mm hole of axis (0, 0, 1), which meets the bracket at (60, 0, 100) mm. The maximum load is (0, 0, -2000) N. It is static.
>
> Requirements: maximum deflection 2 mm, safety factor 2 on yield. Minimise the mass.
>
> Titanium, CNC machined.

Interface	Position (mm)	Axis	Diameter (mm)
bolt hole 1	(0, -40, 200)	(1, 0, 0)	16.0
bolt hole 2	(0, 40, 200)	(1, 0, 0)	16.0
load hole	(60, 0, 100)	(0, 0, 1)	20.0

Requirement	Value
component	Shelf bracket
force	(0, 0, -2000) N
load_type	static
min_safety_factor	2.0
max_displacement	2.0
objective	mass
candidate_materials	['Titanium']
manufacturing	CNC machined

2. Review of the request

The request is complete and consistent.

3. Bracket placement

The bracket is built in its own frame and placed on the requested interface:

Local axis	Global direction
x (load hole to bolts)	(0, 0, 1)
y	(0, -1, 0)
z (out of the support)	(1, 0, 0)
origin	(0, 0, 100)

Load in the bracket frame: (-2000, 0, 0) N.

Parameter fixed by the interface	Value (mm)
bolt_hole_diameter	16.00
bolt_hole_spacing	80.00
bolt_hole_position	100.00
base_length	124.00
load_hole_diameter	20.00
load_hole_height	60.00

Parameter fixed by the interface	Value (mm)
flange_height	90.00

4. Analysis plan

Analysis	Required	Available	Method	Reason
static	yes	yes	linear static FEM, CalculiX C3D10, mesh convergence on the final design	every design is checked for yield and for the displacement limit
fatigue	no	yes	S-N curve with Marin factors, Goodman mean stress correction, Miner damage	the load is static
buckling	no	yes	Euler fixed-free column screening of the flange, factor >= 3, ribs ignored (conservative)	the flange is never in compression
modal	no	no	modal analysis (not implemented)	no vibration excitation was given

5. Solver validation

Case	Quantity	FEM	Reference	Unit	Error	Limit	Result
axial_bar	end displacement	0.047619	0.047619	mm	0.00 %	0.1 %	PASS
axial_bar	axial stress	99.999	100	MPa	0.00 %	0.1 %	PASS
axial_bar	fatigue damage, 1e5 cycles at 200 MPa amplitude, 100 MPa mean	0.13463	0.13465	-	0.01 %	0.1 %	PASS
cantilever	tip deflection	0.95524	0.95981	mm	0.48 %	2.0 %	PASS
cantilever	bending stress at mid span, top fibre	74.999	75	MPa	0.00 %	2.0 %	PASS
plate_with_hole	peak stress at the hole	317.43	313.52	MPa	1.25 %	3.0 %	PASS

6. Material selection

Ashby indices for a plate in bending, minimum mass, normalised by the best candidate. A material scores its worst index.

Material	Score	Limiting	strength	stiffness
Ti-6Al-4V	1.00	strength	1.00	1.00

7. Optimisation

Iteration	Material	Trials	Valid	Best	Smallest margin
1	Ti-6Al-4V	15	8	1004.7 g	14 %
2	Ti-6Al-4V	15	7	960.5 g	7 %

8. Final design

Material: **Ti-6Al-4V**, mass **960.5 g**, indicative cost 1005.07 EUR.

Parameter	Value (mm)
width	145.75
base_length	124.00
base_thickness	7.78
flange_height	90.00
flange_thickness	6.67
fillet_radius	6.74
bolt_hole_diameter	16.00

Parameter	Value (mm)
bolt_hole_spacing	80.00
bolt_hole_position	100.00
load_hole_diameter	20.00
load_hole_height	60.00

Hole	Global position (mm)
bolt_hole_1	(0, 40, 200)
bolt_hole_2	(0, -40, 200)
load_hole	(60, 0, 100)

Result	Value
max displacement	1.86 mm
design von Mises stress	120 MPa
raw peak von Mises (support singularity included)	225.2 MPa
yield safety factor	7.34
mesh	240881 nodes, 147712 C3D10 elements

Checks

Result	Check	Value	Limit	Meaning
PASS	solver_success			ccx finished with result files
PASS	mesh_quality	0.4274	0.1	minimum Gmsh SICN element quality
PASS	load_on_free_dofs	0	0.001	share of the applied load put on constrained dofs and lost to supports
PASS	equilibrium	6.215e-08	0.001	relative difference between reactions and applied loads
PASS	small_displacement	0.008797	0.01	max displacement / part size, linear geometry assumption
PASS	linear_elastic	120	880	design von Mises stress must stay below yield for a linear elastic model
PASS	static_safety_factor	7.336	2	yield strength / design von Mises stress
PASS	max_displacement	1.86	2	
PASS	mesh_convergence_displacement	0.000643	0.02	relative change of max displacement between the two finest meshes
PASS	mesh_convergence_stress	0.009715	0.05	relative change of design von Mises between the two finest meshes

Figures and files

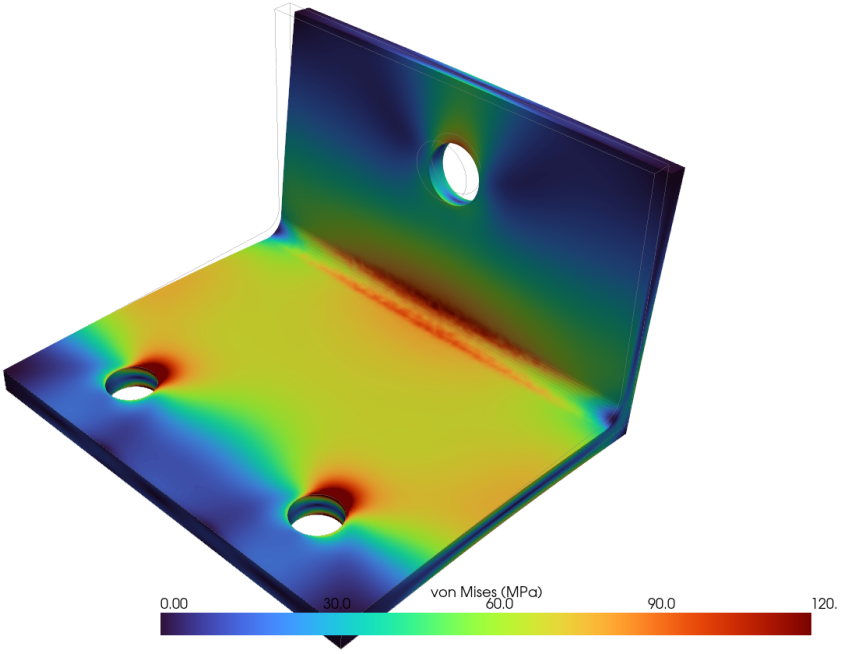
- CAD model in the global frame of the request: `design.step`

Mesh: 240881 nodes, C3D10



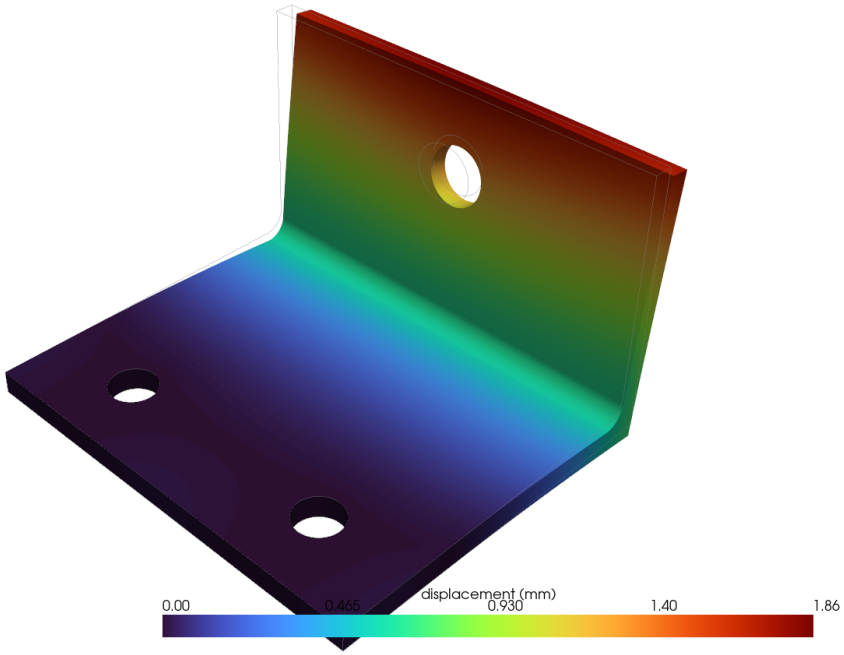
Final mesh (bracket frame)

von Mises (MPa), deformed shape x 6
colour scale capped at the design stress 120 MPa, higher values sit at the rigid bolt supports



von Mises stress on the deformed shape (bracket frame)

displacement (mm), deformed shape x 6



Displacement on the deformed shape (bracket frame)

9. Assumptions

- Linear elastic isotropic material.
- Small displacements and strains (geometrically linear), static loading.
- Quadratic tetrahedral elements (CalculiX C3D10) meshed with Gmsh.
- Nodal stresses are extrapolated from integration points and averaged by CalculiX.
- Design stresses ignore support nodes and nodes within the stress exclusion distance.
- Units: mm, N, MPa, tonne, s.
- Bolt holes are fully fixed: rigid bolts, no preload, no contact or friction with the support surface. This is stiffer than a real bolted joint and gives a stress singularity at the hole edges, so stresses within a quarter of the bolt hole diameter of the holes are excluded from the design values.
- The load is introduced by a rigid pin: the total force is applied as a uniform traction on the half of the load hole facing the force direction (no contact analysis).

10. Decision trace

1. [llm] **parse**: request extracted from the text (every extracted number and vector is checked against the text)

2. [rule] **plan**: static: required (every design is checked for yield and for the displacement limit. Method: linear static FEM, CalculiX C3D10, mesh convergence on the final design)

3. [rule] **plan**: fatigue: not required (the load is static)

4. [rule] **plan**: buckling: not required (the flange is never in compression)

5. [rule] **plan**: modal: not required (no vibration excitation was given)

6. [tool] **validate_solver**: 6 of 6 analytical reference cases passed (results are only trusted once the solver chain reproduces known solutions)

7. [tool] **select_materials**: Ranked 1 materials on strength, stiffness for minimum mass (score, limiting index): Ti-6Al-4V (1.00, limited by strength)

8. [rule] **select_materials**: shortlist: Ti-6Al-4V (the 2 best ranked materials are optimised)

9. [tool] **search 1**: Ti-6Al-4V: best valid design 1004.7 g (8 valid designs out of 15 trials). Closest requirement: max_displacement with 14 % margin. No requirement is near its limit, so the search has probably not converged: more trials should give a better design.

10. [tool] **verify 1**: Chose Ti-6Al-4V (1004.7 g), the lightest valid design that passed the final verification: mesh convergence and every check on the finer mesh.

11. [tool] **verify 1**: Closest requirement: max_displacement with 14 % margin. No requirement is near its limit, so the search has probably not converged: more trials should give a better design.

12. [llm] **decide 1**: refine (allowed: refine, accept) (Since no requirement is within 10% of its limit, the current design is not near the Pareto frontier, indicating that further trials are likely to yield a lighter design.)

13. [rule] **decide 1**: refine Ti-6Al-4V within +/- 30 % of their best design, starting from it

14. [tool] **search 2**: Ti-6Al-4V: best valid design 960.5 g (7 valid designs out of 15 trials). Closest requirement: max_displacement with 7 % margin.

15. [tool] **verify 2**: Chose Ti-6Al-4V (960.5 g), the lightest valid design that passed the final verification: mesh convergence and every check on the finer mesh.

16. [tool] **verify 2**: Closest requirement: max_displacement with 7 % margin.

17. [rule] **decide 2**: accept (allowed: accept) (the only action the rules allow)

11. Limitations

- Bolt holes are modelled as rigidly fixed: no preload, contact or friction. Stresses within a quarter of the bolt hole diameter of the holes are excluded from design values, and fatigue of the bolted joint itself is not assessed.
- The pin load is a uniform traction on the bearing half of the hole, without contact.
- Linear elastic, small displacements. Stress-life fatigue only, no notch sensitivity reduction (conservative), no crack growth.
- Buckling is screened with a conservative Euler column, there is no buckling FEM.
- Material properties, S-N data and prices are typical handbook values, not certified design allowables.
- The optimisation samples the design space: the result is a good verified design, not a proven global optimum.